# BCH Codes: Encoding and Decoding Process

Jimmy Sitompul[1] and Michael Liu[2]

[1]Department of Mathematics, Colorado State University
[2]Department of Electrical and Computer Engineering, Colorado State University

## 1 Introduction

In terminology, a BCH (Bose-Chaudhuri-Hocquenghem) code is a variable-length digital encoding that is multi-level, cyclic, error-correcting, and used to correct a variety of random error patterns at the same time. It is also possible to use BCH codes for multi-level phase-shift keying of prime numbers in either the prime order or the power order of prime numbers. There are three individual discoveries of this binary linear cyclic code in 1959: those such as Bose, Chaudhuri, and Hocquenghem. It was given the name BCH code, which was derived from the prefix of their names.

One of the important characteristics of BCH codes is that the number of symbol errors that the code can fix may be precisely controlled during code design. It is possible to create binary BCH codes, in particular, that can fix multiple bit errors. BCH codes also have the advantage of being easy to decode using an algebraic technique called syndrome decoding, which is also called minimum distance decoding. A few examples of real-world applications using BCH codes include satellite communications, CD/DVD players, disk drives, USB flash drives, solid-state drives, quantum-resistant cryptography, and two-dimensional bar codes.

## 2 Definition of BCH Codes

Let $p(x)$ be a degree $r$ primitive polynomial with binary field coefficients. The polynomial $a(x)$ is called a $t$-error correcting BCH code if $a(x)$ is a non-zero polynomial with the properties

$$a(x) = a(x^3) = a(x^5) = ... = a(x^{2t-1}) = 0 \quad (mod \;\; p(x))$$

for all $t$ such that $1 \leq 2t - 1 \leq 2r - 1$. Furthermore, for any positive integers $m \geq 3$ and $t < 2m - 1$, there exists a binary BCH code with the following parameters:

1. Block length: $n = 2m - 1$
2. Minimum distance: $d \geq 2t + 1$
3. Number of parity-check digits: $n - k \leq mt$

BCH code is a block code, its theory is relatively complete, but the coding method is more complicated. The next sections explain how the encoding and decoding processes of BCH codes work. It is necessary to first discuss the generator polynomial of a BCH code, which is discussed in the following section.

## 3 Generator Polynomials of BCH Codes

The generator polynomial of the BCH code is the product of the lowest common multiple (LCM) of $2t$ minimum polynomials. The generator polynomial of the BCH code with error correction capability is $g(x)$.

$$g(x) = LCM[p(x), p_3(x), ..., p_{2t-1}(x)]$$

where $p(x), p_3(x), ..., p_{2t-1}(x)$ are minimal polynomials. In addition, the following condition has to be satisfied

$$p_3(x^3) = p_5(x^5) = ... = p_{2t-1}(x^{2t-1}) = 0 \quad (mod \;\; p(x))$$

or in other words, the polynomial $p(x)$ divides all the polynomials $p_3(x^3), ..., p_{2t-1}(x^{2t-1})$. Given a set of unique minimum polynomials, a BCH generating polynomial can generate codewords with predictable distance properties. Let $t$ be the total amount of bit errors that need to be corrected. $d = 2t + 1$ is the necessary Hamming

distance. Find a set of minimum polynomials for $n = 1$ to $t$ from $a^n$. Eliminate any minimum polynomials that are duplicates to reduce the set. The product of the remaining minimal polynomials yields the BCH generating polynomial $g(x)$.

# 4 Encoding

Because any polynomial that is a multiple of the generator polynomial $g(x)$ is a valid BCH codeword, the process of selecting a polynomial that has the generator polynomial as a factor is known as BCH encoding. Therefore, the transmitted codeword $s(x)$ is of the form

$$s(x) = p(x)g(x)$$

where $p(x)$ is the message that we want to encode and this refers to the non-systematic encoding of BCH codes.

## Example

Suppose that we have the generator polynomial $g(x) = x^{10}+x^9+x^8+x^6+x^5+x^3+1$ in the (31,21) binary BCH code and that we want to encode a 21-bit message {101101110111101111101}, we need to represent the 21-bit message as a polynomial $p(x)$:

$$p(x) = x^{20} + x^{18} + x^{17} + x^{15} + x^{14} + x^{13} + x^{11} + x^{10} + x^9 + x^8 + x^6 + x^5 + x^3 + 1$$

As a result, the transmitted codeword $s(x)$ is

$$s(x) = (x^{20} + x^{18} + x^{17} + x^{15} + x^{14} + x^{13} + x^{11} + x^{10} + x^9 + x^8 + x^6 + x^5 + x^3 + 1)(x^{10} + x^9 + x^8 + x^6 + x^5 + x^3 + 1)$$

Therefore, $s(x) = x^{30}+x^{29}+x^{26}+x^{25}+x^{24}+x^{22}+x^{19}+x^{17}+x^{16}+x^{15}+x^{14}+x^{12}+x^{10}+x^9+x^8+x^6+x^5+x^4+x^2+1$ and the transmitted codeword can be written as the binary code {1100111010010111101011101110101}.

## Euclidean Algorithm in Encoding Process

This refers to the systematic BCH encoding which involves taking the message polynomial $p(x)$ and multiply it by $x^{n-k}$ and then use the Euclidean algorithm to obtain:

$$p(x)x^{n-k} = q(x)g(x) + r(x)$$

where $r(x)$ is the remainder polynomial.
The remainder polynomial $r(x)$ has a degree less than $g(x)$, in other words, $deg(r(x)) < deg(g(x)) = n - k$. Here, we can subtract the remainder polynomial r(x) from $p(x)x^{n-k}$ without altering any of the message coefficients, so we obtain:

$$s(x) = q(x)g(x) = p(x)x^{n-k} - r(x)$$

Here, $q(x)g(x)$ is a valid codeword.

# 5 Decoding

According to the generator polynomial, a fast hardware encoder can be constructed, and for the decoding of BCH codes, since it is a subclass of cyclic codes, any standard decoding process for cyclic codes is applicable to BCH codes. A more efficient algorithm for BCH codes is the Gorenstein-zierler decoding algorithm. Let $s(x)$ be the transmitted codeword polynomial and $e(x)$ be the error polynomial, then the received polynomial is $r(x) = s(x) + e(x)$. Let $y_1, y_2, ..., y_w$ be the root of $g(x)$ on $GF(p^m)$, i.e. $g(y_i) = 0$, $i = 1, 2, ..., w$. Because for a certain information polynomial $a(x)$, there is

$$s(x) = a(x)g(x)$$

So,

$$s(y_i) = 0, r(y_i) = s(y_i) + e(y_i) = e(y_i), \quad i = 1, 2, ..., w.$$

Assuming that the BCH code is constructed from a field element $a$, consider the error polynomial $e(x) = e_{n-1}x^{n-1}+e_{n-2}x^{n-2}+...+e_1x+e_0$. Among them, there are at most t coefficients that are non-zero ($t$ errors can be corrected), assuming that $v$ errors have actually occurred, where $0 \leq v \leq t$. Assuming that the error occurs at positions $i_1, i_2, ..., i_v$, the error polynomial can be written as

$$e(x) = e_{i_1}x^{i_1} + e_{i_2}x^{i_2} + ... + e_{i_v}x^{i_v}$$

where $e_{i_k}$ is the size of the $k$-th error. For binary codes, $e_{i_k} = 1$.

For decoding steps of BCH code, first, as a test value, let $v = t$ and calculate the determinant of the adjoint matrix $M$. If the value of the determinant is zero, let $v = t-1$ and compute the determinant of $M$ again. This process is repeated until a value of $v$ is found such that the determinant of the adjoint matrix is not 0, and the value of $v$ is the actual number of errors. Second, find the inverse of $M$ and compute the coefficients of the error location polynomial $U(x)$. Third, solve for the zero point of $U(x) = 0$, from which the error positions $X_1, X_2, ..., X_v$ can be calculated. If it's a binary code, stop here (because the error level is 1). Last, if it is not a binary code, go

back to the system of equations and solve these equations to get the error degree.

$$S_1 = Y_1X_1 + Y_2X_2 + ... + Y_vX_v$$

$$S_2 = Y_1X_1^2 + Y_2X_2^2 + ... + Y_vX_v^2$$

$$S_{2t} = Y_1X_1^{2t} + Y_2X_2^{2t} + ... + Y_vX_v^{2t}$$

# 6 Conclusions

BCH code is a cyclic error-correcting code that can be built over a finite field. There are several advantages to implementing BCH codes, including the fact that they can be decoded using syndrome decoding, which is extremely fast, the fact that they are commonly used, and the fact that they are flexible in that they offer flexibility over block length. Despite the complexity of the decoding algorithm, BCH codes are still commonly employed in real-world applications such as DVDs, disk drives, satellite communications, and other similar technologies.

# 7 References

Han, Yunghsiang S. BCH Codes. Taiwan National Taipei University, https://web.ntpu.edu.tw/~yshan/BCH_code.pdf#:~:text=Description%20of%20BCH%20Codes%20%E2%80%A2%20The%20Bose%2C%20Chaudhuri%2C,generalization%20of%20the%20Hamming%20code%20for%20multiple-error%20correction.

Ho, Shan-Yuan. BCH Codes. Massachusetts Institute of Technology, 24 Apr. 2010. Wikipedia Contributors. "BCH Code." Wikipedia, Wikimedia Foundation, 8 Feb. 2019, https://en.wikipedia.org/wiki/BCH_code.